

Diverse Evolving Worlds

A Standard Simulation Worlds

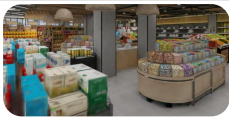


HM3D



MP3D

B Photorealistic Simulation



Habitat-GS



Interior-GS

C Evolving Simulation Worlds



HSSD

D Real-World Evolution



Indoor



Outdoor

Memory
Construction

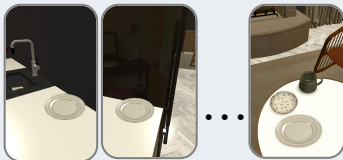


EvolvingNav



Memory
Update

1 Observed History



t_1

t_2

t_k

2 Hidden Evolution



world evolves

3 Current Belief



state₁

state₂

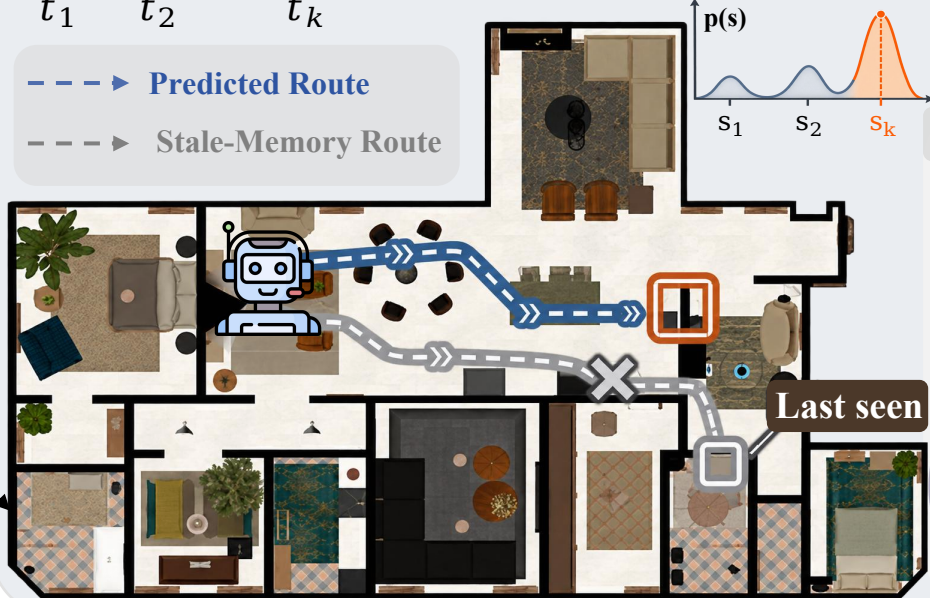
state_k

4 Belief-guided Plan/Act



---> Predicted Route

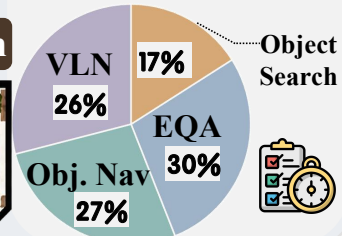
---> Stale-Memory Route



EvoWorld-Bench



54 Scenes
803.68k Tasks
4860 Scene-Days
239.19k Dynamic State Changes



Unified Embodied Tasks

A Embodied QA

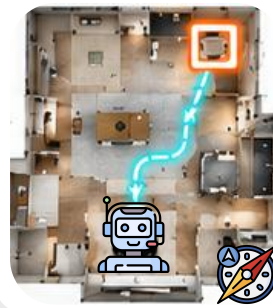


Q: Where did I put the cup after I left the dining table ?

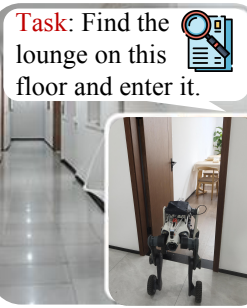
B Active Exploration



C Predictive Navigation



D Real World Interaction



Task: Find the lounge on this floor and enter it.